

Photoaging and Matrix Degradation

UV radiation aggravates the degradation of the dermal matrix by increasing levels of **MMP-1** and **MMP-9**. Matrix degradation is further exacerbated by **MMP-8** (collagenase) of neutrophil origin following neutrophil infiltration into UV-irradiated skin. Although there is a concomitant upregulation of **tissue inhibitors of metalloproteinases (TIMPs)** that limit matrix degradation, TIMPs are presumably not fully effective in preventing cumulative damage to dermal collagen.

UV-induced collagen degradation is typically incomplete, leading to the accumulation of partially degraded collagen fragments in the dermis, which compromises the structural integrity of the skin. Additionally, these large collagen degradation products inhibit new collagen synthesis, meaning that collagen degradation itself negatively regulates collagen production. Interestingly, increased stress-associated **AP-1 activity**, elevated **CYR61**, and higher **MMP levels**, along with reduced collagen production, are also observed in intrinsically aged skin, suggesting that similar mechanisms—possibly driven by **ROS generation**—contribute to chronological aging, as discussed in the section on Mechanisms of Intrinsic Skin Aging.

Mitochondrial Damage

Mitochondria are organelles responsible for energy production in the form of **adenosine triphosphate (ATP)** through oxygen consumption. Despite having antioxidant defense systems, mitochondria are susceptible to continuous **ROS generation**, which damages **mitochondrial DNA (mtDNA)**. To date, no machinery for removing bulky DNA lesions has been identified in mitochondria, although they possess base excision repair mechanisms effective against oxidative damage. Nevertheless, mtDNA mutation frequency is approximately **50-fold higher** than that of nuclear DNA. Photodamaged skin exhibits even higher mtDNA mutation rates compared to sun-protected skin, including large DNA deletions, resulting in impaired mitochondrial function.

This leads to further ROS accumulation and reduced cellular energy production.

A correlation exists between decreased mitochondrial function and increased **MMP-1** levels without a corresponding rise in the specific TIMP for MMP-1, thereby exacerbating collagen degradation and accelerating skin photoaging.

Protein Oxidation

Oxidative damage affects proteins, and photodamaged skin shows accumulation of oxidized, dysfunctional proteins in the upper dermis. In vitro studies indicate that **UVA** is a major contributor to this effect. The buildup of oxidized proteins further impairs **proteasomal function**, reducing the cell's ability to clear additional damaged proteins.

Action Spectrum of Photoaging

The precise action spectrum for human photoaging remains undetermined. However, studies using animal models have investigated the contributions of different solar spectral bands. In rodent skin, prolonged intense irradiation with either predominantly **UVB** or **UVA** can induce an elastosis-like condition.

UVB photons are, on average, **1,000 times more energetic** than UVA photons and are primarily responsible for sunburn, tanning, and photocarcinogenesis. UVB causes direct **DNA damage**, induces **inflammation** and **immunosuppression**, and stimulates the synthesis and release of **prostaglandins**, particularly **PGE₂**, via induction of **cyclooxygenase-2 (COX-2)**. UVB also activates **ornithine decarboxylase**, the rate-limiting enzyme in polyamine biosynthesis, promoting cellular proliferation (contributing to carcinogenesis) and cutaneous angiogenesis by downregulating the angiogenesis inhibitor **thrombospondin-1** and upregulating **VEGF** and **platelet-derived endothelial cell growth factor**. Additionally, UVB promotes the infiltration of elastase-producing leukocytes, worsening elastin degradation.

Despite the potent biological effects of UVB, **UVA** is believed to play a disproportionately larger role in photoaging due to its: - At least **10-fold greater abundance** in terrestrial sunlight, - Higher year-round and daily average irradiance, - Deeper penetration into the dermis compared to UVB.

Notably, human skin exposed daily for just one month to **sub-erythemogenic doses of UVA** alone develops: - Epidermal hyperplasia, - Stratum corneum thickening, - Langerhans cell depletion, - Dermal inflammatory infiltrates, - Lysozyme deposition on elastic fibers.

UVA also induces the synthesis and release of **cytokines** and **MMPs**, particularly **MMP-1 (collagenase)** and **elastase**, and triggers **mtDNA mutations**, all of which contribute significantly to skin photoaging.